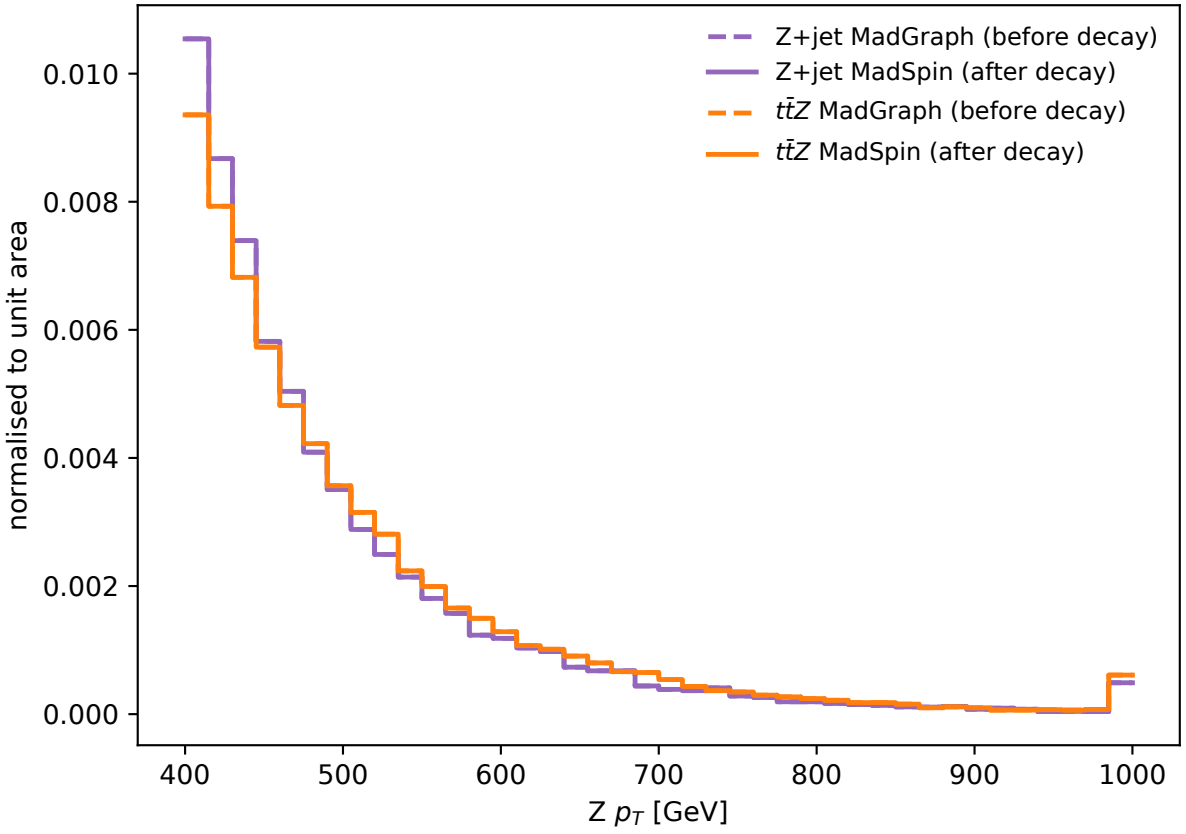
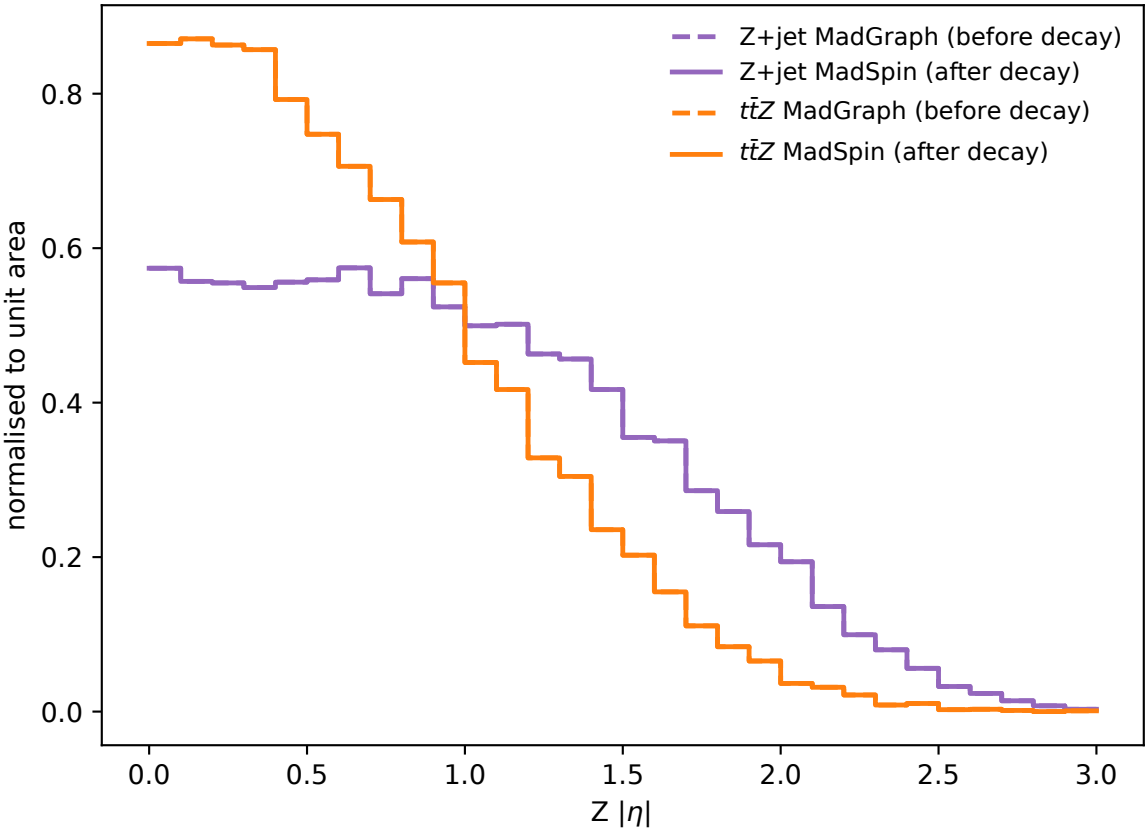
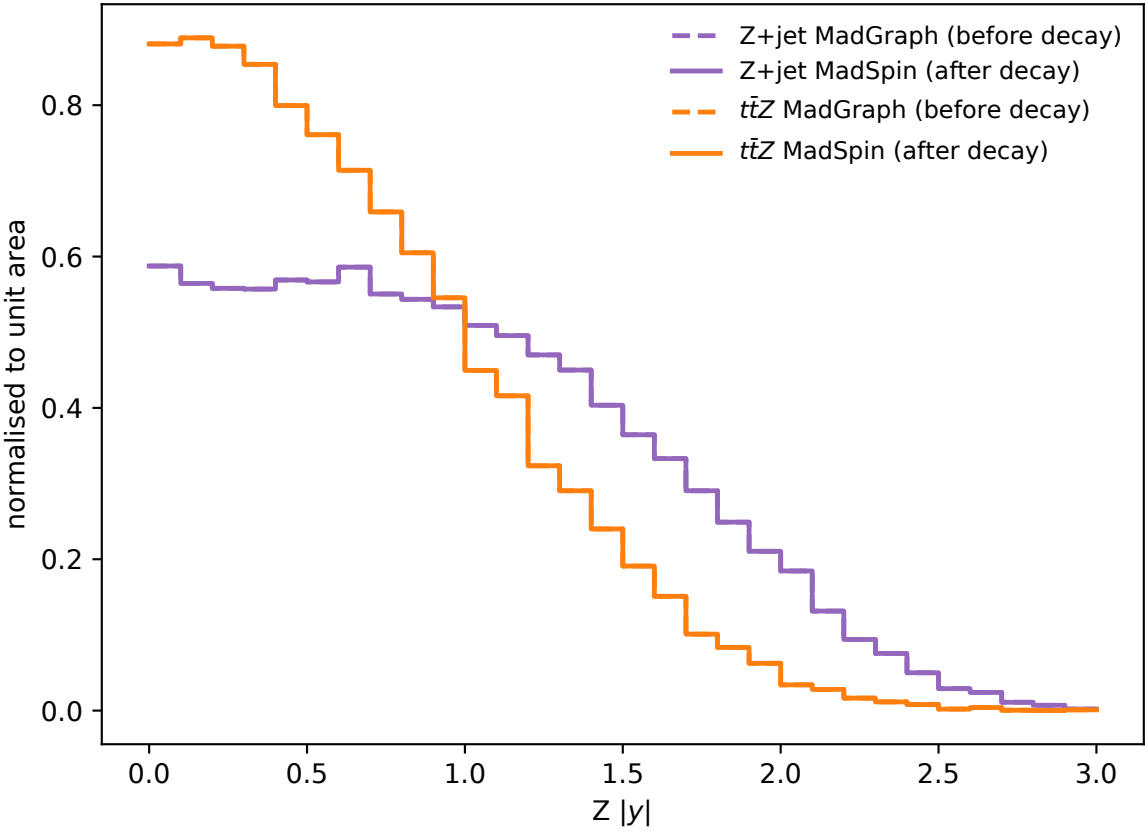
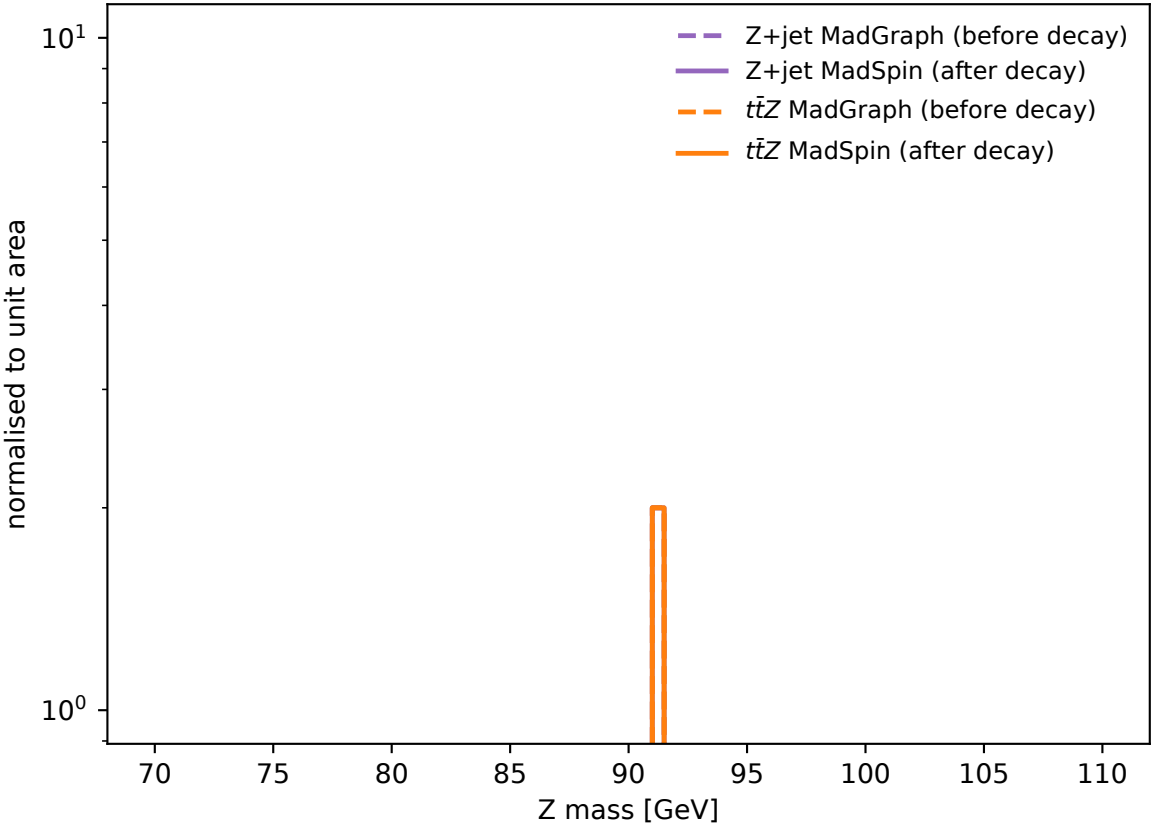
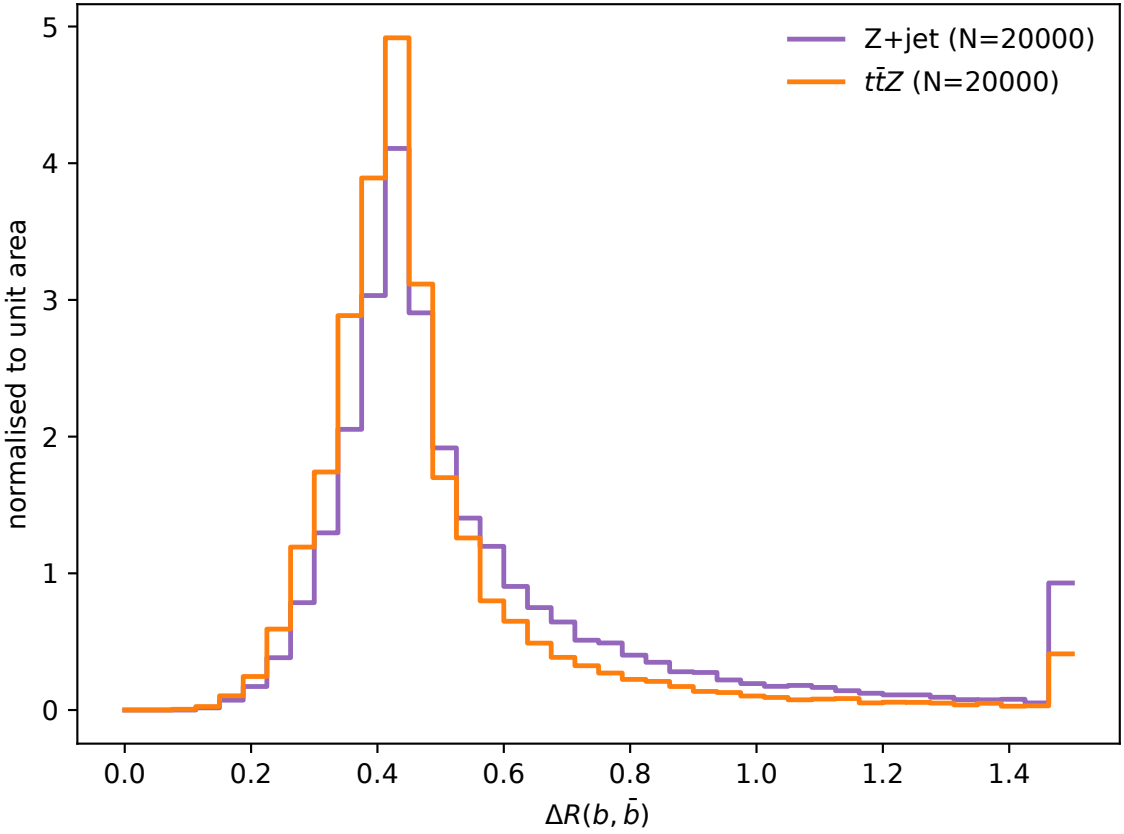
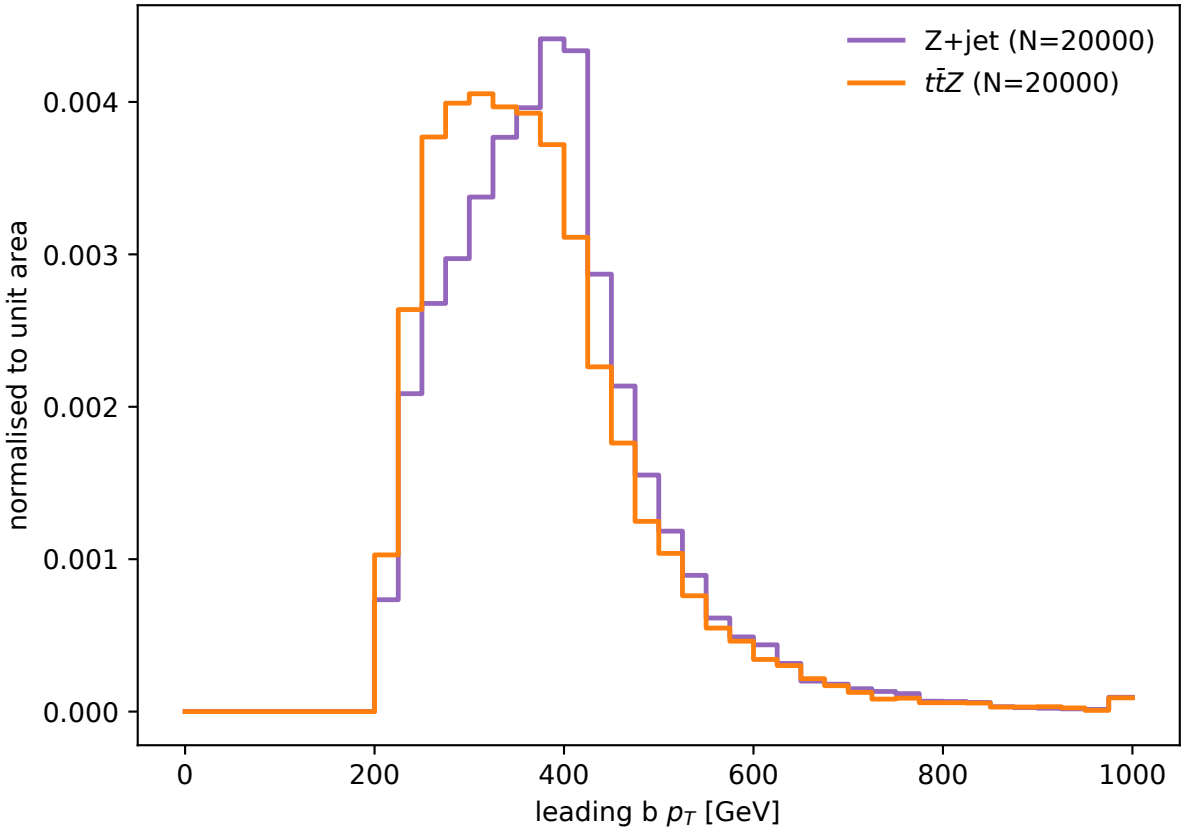

Z+jet MadGraph (before decay)
Z+jet MadSpin (after decay)
$t\bar{t}Z$ MadGraph (before decay)
$t\bar{t}Z$ MadSpin (after decay)
normalised to unit area
0.010
0.008
0.006
0.004
0.002
0.000
400
500
600
700
800
900
1000
Z p_T [GeV]

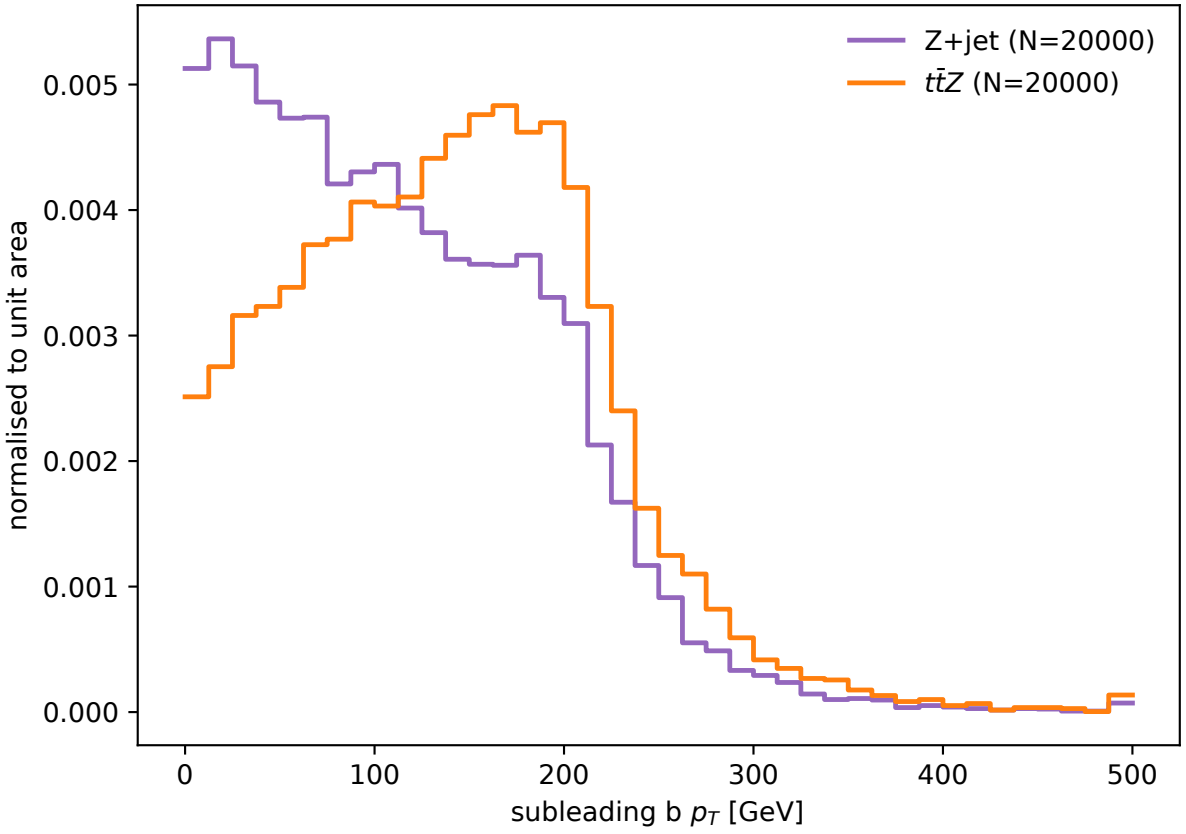

Z+jet MadGraph (before decay)
Z+jet MadSpin (after decay)
$t\bar{t}Z$ MadGraph (before decay)
$t\bar{t}Z$ MadSpin (after decay)
normalised to unit area
0.0
0.2
0.4
0.6
0.8
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Z $|\eta|$

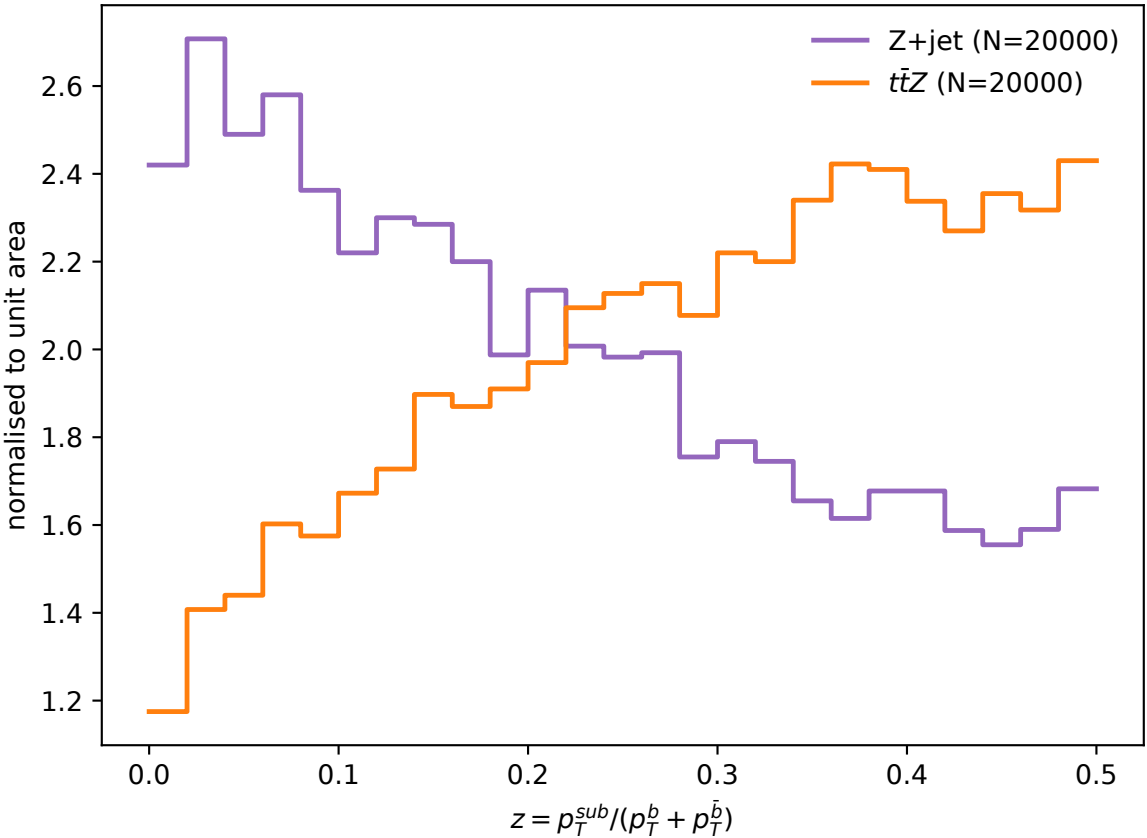

Z+jet MadGraph (before decay)
Z+jet MadSpin (after decay)
$t\bar{t}Z$ MadGraph (before decay)
$t\bar{t}Z$ MadSpin (after decay)
normalised to unit area
0.0
0.2
0.4
0.6
0.8
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Z $|y|$

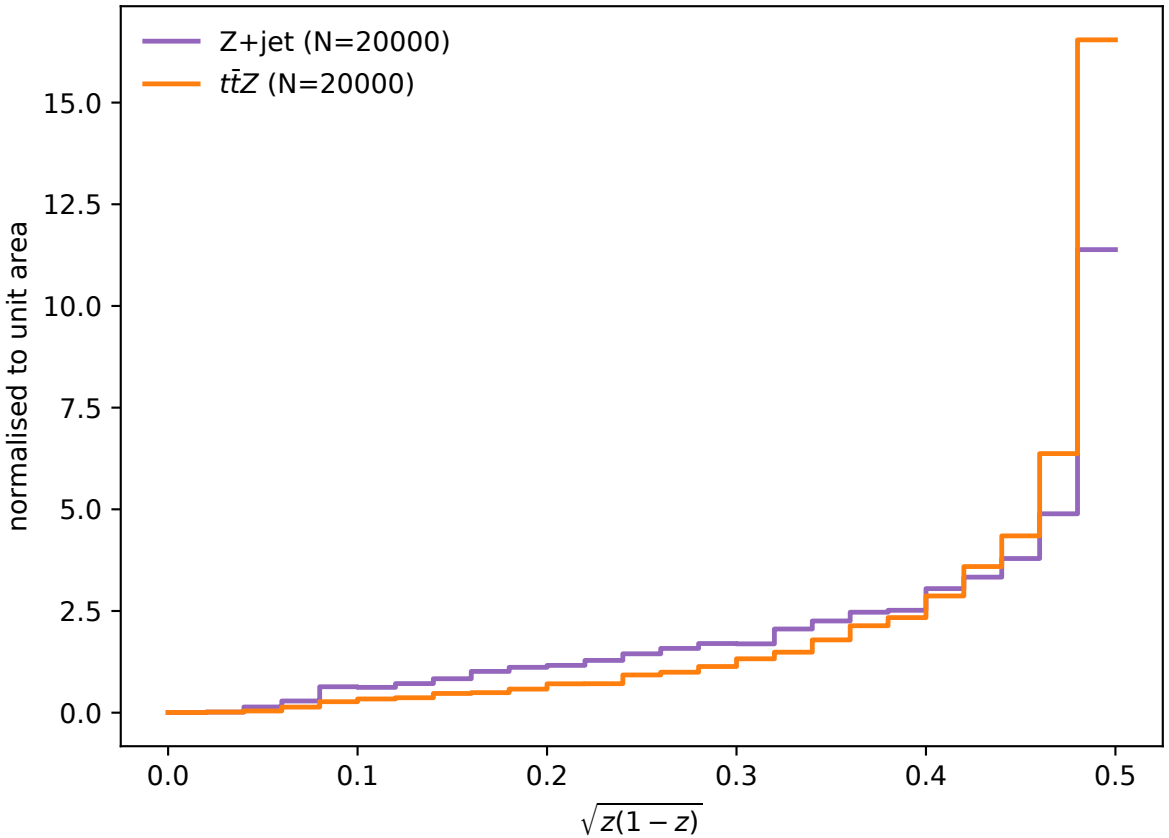

Z+jet MadGraph (before decay)
Z+jet MadSpin (after decay)
$t\bar{t}Z$ MadGraph (before decay)
$t\bar{t}Z$ MadSpin (after decay)
10^1
10^0
normalised to unit area
70
75
80
85
90
95
100
105
110
Z mass [GeV]

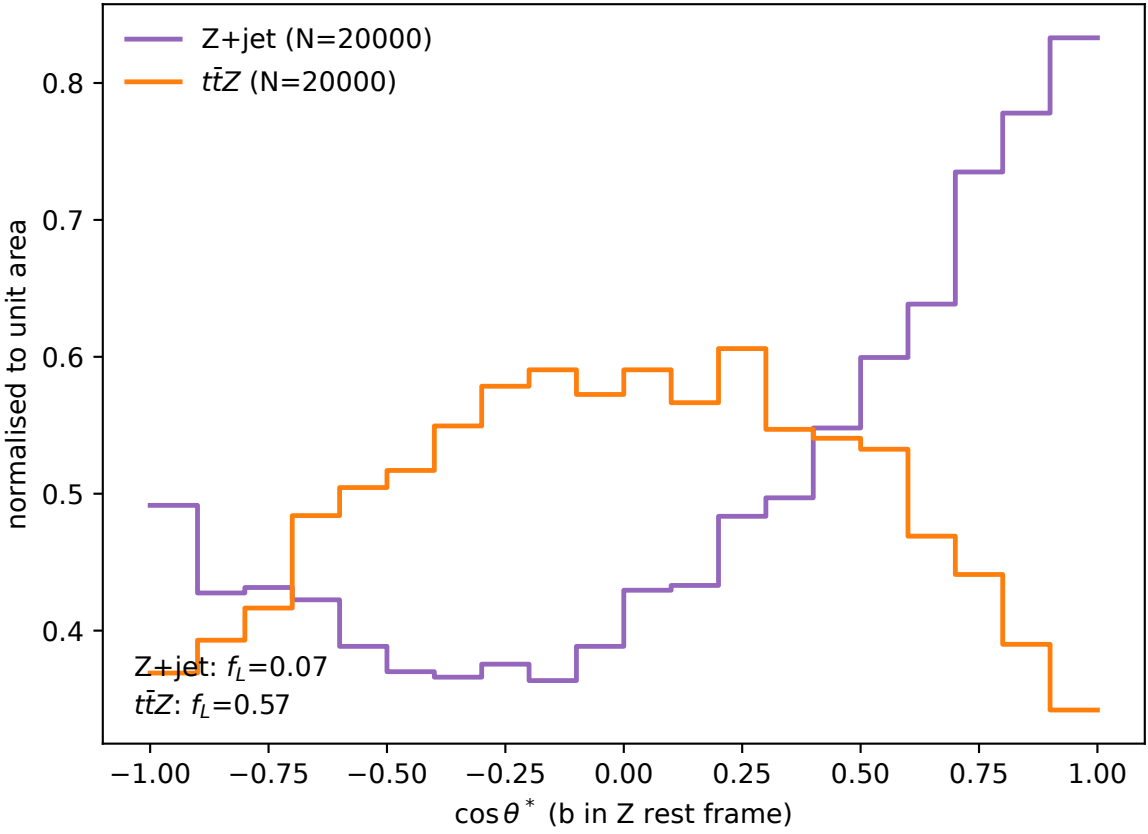

Z+jet (N=20000)
$t\bar{t}Z$ (N=20000)
normalised to unit area
0
1
2
3
4
5
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
$\Delta R(b, \bar{b})$


Z+jet (N=20000)
$t\bar{t}Z$ (N=20000)
normalised to unit area
0.000
0.001
0.002
0.003
0.004
0
200
400
600
800
1000
leading b p_T [GeV]


Z+jet (N=20000)
$t\bar{t}Z$ (N=20000)
normalised to unit area
0.000
0.001
0.002
0.003
0.004
0.005
0
100
200
300
400
500
subleading b p_T [GeV]


Z+jet (N=20000)
$t\bar{t}Z$ (N=20000)
normalised to unit area
2.6
2.4
2.2
2.0
1.8
1.6
1.4
1.2
0.0
0.1
0.2
0.3
0.4
0.5
$z = p_T^{sub}/(p_T^b + p_T^{\bar{b}})$


Z+jet (N=20000)
$t\bar{t}Z$ (N=20000)
normalised to unit area
0.0
2.5
5.0
7.5
10.0
12.5
15.0
0.0
0.1
0.2
0.3
0.4
0.5
$\sqrt{z(1-z)}$


Z+jet (N=20000)
$t\bar{t}Z$ (N=20000)
normalised to unit area
0.8
0.7
0.6
0.5
0.4
Z+jet: f_L=0.07
$t\bar{t}Z$: f_L=0.57
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
$\cos\theta^*$ (b in Z rest frame)